

Workshop No. 1

Student Mental Health Support Platform



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Team Members

Alan Santiago Agudelo Sarmiento – 20222020170

Juan David Cardozo Trujillo - 20231020155

Julian Ernesto Romero Gutierrez - 20231020164

Nicolas Alexander Sierra Contreras - 20222020197

Professor

Eng. Carlos Andrés Sierra, M.Sc.

Universidad distrital Francisco José de Caldas

Engineering faculty

Systems Analysis & Design

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1. System Overview and Scope Definition

1.1 System Description

The Student Mental Health Support Platform is a digital system designed to address the growing mental health challenges faced by university students. The platform operates as a structured peer support network that connects students experiencing emotional or psychological difficulties with trained peer counselors, professional advisors, and curated mental health resources.

The system is composed of three core functional modules: a resource matching engine, an appointment scheduling module, and an anonymous peer counselor connection service. These modules work together to provide students with timely, accessible, and confidential support, while ensuring user experience quality and data privacy compliance.

The platform does not replace professional clinical services but serves as a first point of contact — a bridge between students in distress and the institutional resources available to them. The architecture prioritizes simplicity, scalability, and user trust.

1.2 System Purpose and Context

Mental health issues among university students have increased significantly in recent years. Many students avoid seeking help due to stigma, lack of awareness of available resources, or difficulty navigating institutional processes. This platform aims to lower the barrier to access by offering a familiar, digital-first experience that guarantees user anonymity.

Within the university ecosystem, the platform operates as a complementary service to the existing counseling department. It does not store sensitive clinical records; instead, it focuses on connecting users with the right resource or person at the right time, using simple matching logic based on student-reported needs.

1.3 System Boundaries

The following boundaries define what is inside and outside the scope of this system:

IN SCOPE	OUT OF SCOPE
Student registration and anonymous profile creation	Clinical diagnosis or therapy sessions
Resource matching based on reported needs	Storage of clinical or medical records
Appointment scheduling with counseling services	Integration with hospital or external health systems
Anonymous peer counselor chat connections	Emergency psychiatric intervention services
User experience interface (web/mobile)	Payroll or administrative HR systems
Data privacy and consent management	Academic performance tracking

1.4 Operating Environment

The platform is designed to operate within the university's digital ecosystem. It is accessible via web browser and mobile devices, requiring only an internet connection and a university-issued student account for authentication. The system runs on a cloud-based server environment, ensuring continuous availability and secure access to user data.

The platform handles moderate concurrent user loads, with peak usage expected during midterm and final exam periods. It is designed to remain functional with minimal maintenance overhead, using standard web technologies to ensure long-term supportability.

1.5 Stakeholder Identification

The following stakeholders are directly or indirectly involved in the system:

Stakeholder	Role	Interest in the System
Students	Primary Users	Access to mental health resources and anonymous peer support
Peer Counselors	Support Providers	Provide support through anonymous connections
Professional Counselor	Counseling Service Provider	Receive appointment requests from students and manage scheduled counseling sessions through the platform
University Admin	System Supervisors	Ensure compliance, data privacy, and service quality
Platform Admins	Technical Operators	Maintain system availability, matching algorithms, and data integrity

Each stakeholder group interacts with the system in different ways. Students are the primary users and the central focus of the platform's design. Peer counselors serve as volunteer support providers who are trained and verified before joining the platform. Professional counselors are the university therapists and psychologists who receive and attend the appointments scheduled through the platform. University administrators oversee the system's alignment with institutional policies and define the professional counselors available on the platform. Platform administrators handle the technical operation and integrity of the system.

2. Data Collection Plan and Execution

2.1 Methodology Design

The data collection strategy for this project was designed to understand students' mental health experiences and evaluate the potential usefulness of a **Student Mental Health Support Platform**. Three primary data collection methods were considered: **Structured Surveys, Usage Monitoring, and Process Documentation**.

The main method used in this study was a **structured online survey** distributed through a digital form to university students. The survey collected information about students' stress levels, access to mental health services, willingness to seek support, and interest in using a digital platform for mental health assistance.

The questionnaire consisted of **10 multiple-choice questions**, allowing responses to be easily analyzed and summarized statistically. The survey was distributed through online channels such as messaging platforms and student groups. The target population consisted of university students, as they represent the primary users of the proposed system. A sample of approximately **50 participants** was expected for the study.

In addition to surveys, the methodology conceptually incorporates **usage monitoring**, which would allow tracking user interactions with the platform. This includes monitoring the frequency of platform access, the use of peer-support features, and the interaction with educational mental health resources.

The third method considered was **process documentation**, which describes the workflow of the proposed system. This process includes user registration, browsing available mental health resources, connecting with peer-support networks, and requesting anonymous counseling. Documenting these processes helps analyze how users would interact with the system and identify potential improvements in its design.

The data collection process was conducted over a period of approximately one week, allowing sufficient time to distribute the survey and collect responses.

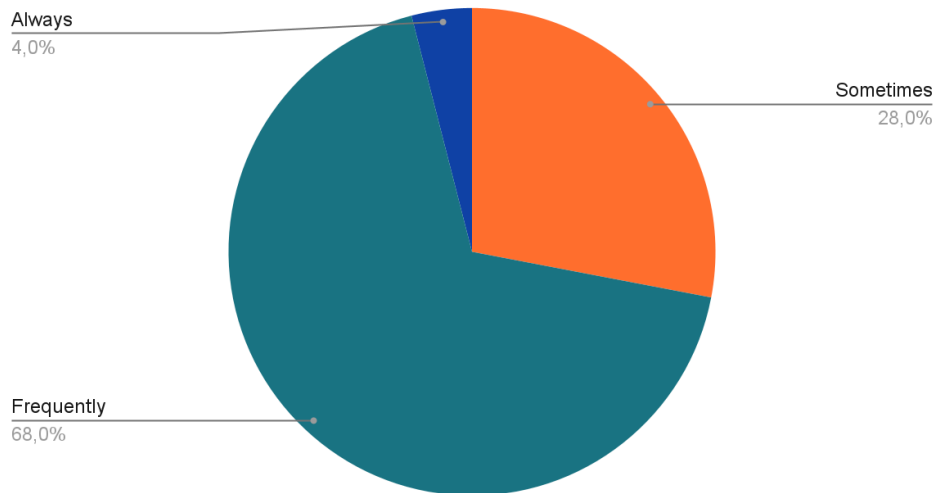
2.2 Primary Data Presentation

The primary data collected from the survey responses was organized and analyzed using tables and charts in order to identify patterns related to student mental health and support needs.

Academic Stress Levels

Stress Level	Students
Never	0
Rarely	0
Sometimes	7
Frequently	17
Always	1

Academic Stress Levels

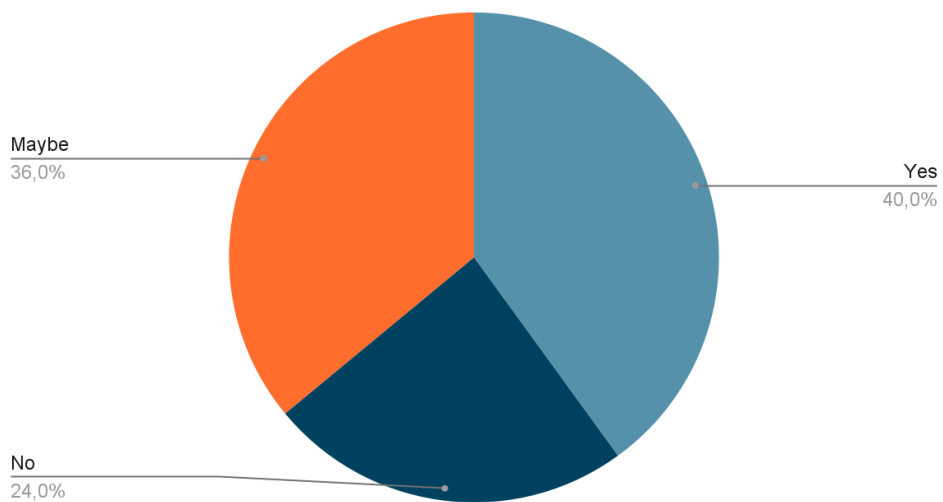


These results indicate that most students frequently experience stress during their academic life, highlighting the need for accessible mental health support resources.

Interest in Using a Mental Health Support Platform

Response	Students
Yes	10
No	6
Maybe	9

Interest in Using a Mental Health Support Platform

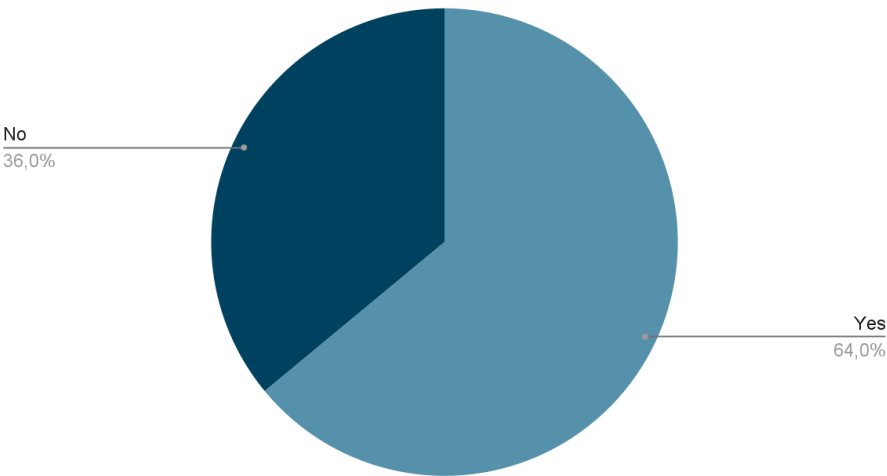


The results show that a majority of students would be willing to use a digital platform to receive mental health support.

Preference for Anonymous Support

Response	Students
Yes	16
No	9

Preference for Anonymous Support

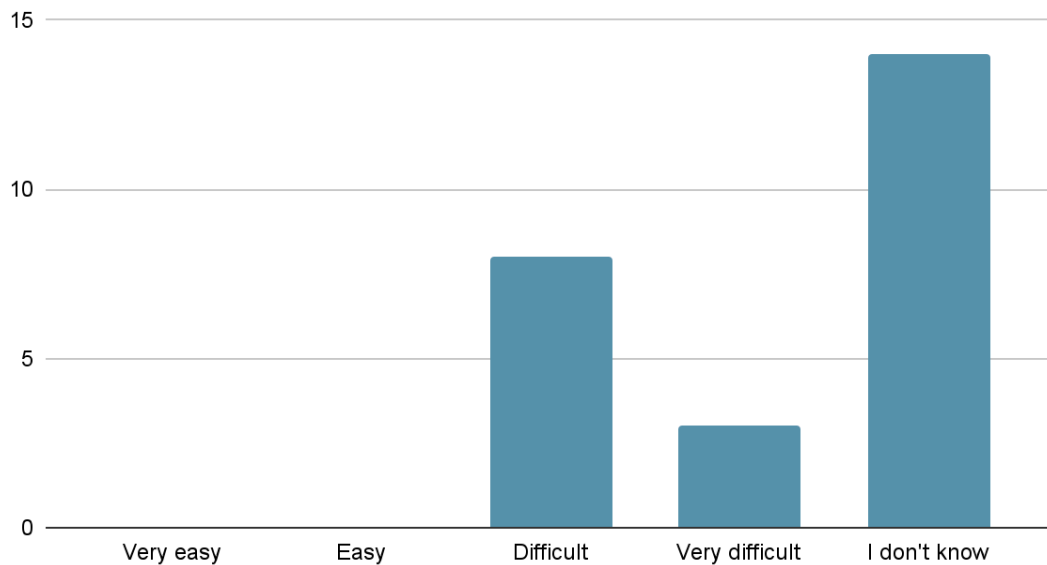


This suggests that anonymity plays an important role in encouraging students to seek help, which should be considered when designing the platform.

How easy is it to access mental health services in the university?

Response	Students
Very easy	0
Easy	0
Difficult	8
Very difficult	3
I don't know	14

How easy is it to access mental health services in the university?



Most people respond that it is difficult to access mental health services or they are unaware of the services, which should be taken into account when developing the platform, ensuring that it is easily accessible to students and that most people are aware of its existence.

To improve the **interpretation of the data**, the results were also visualized using bar charts and pie charts, which allow trends and response distributions to be easily identified.

The following data were collected from the survey responses but are not considered primary data.

Have you ever experienced anxiety or emotional problems during your university life?

Response	Students
Yes	22
No	2

Have you ever sought help for mental health concerns?

Response	Students
Yes	8
No	17

How comfortable would you feel discussing your mental health with other students?

Response	Students
Yes	4
No	8
Maybe	13

What type of support would you like to find on a mental health platform?

Response	Students
Professional counseling	8
Peer support	0
Educational mental health resources	0
All of the above	17

How often do you think you would use a mental health support platform if it were available?

Response	Students
Never	6
Occasionally	13
A few times per month	5
Frequently	1

Do you believe that a peer-support platform could improve student mental health?

Response	Students
Yes	15
No	2
Maybe	8

These responses can help us model some aspects we need on the platform and that's why they were included in the form, but they are not considered primary data.

2.3 Data Quality Assessment

Although the collected data provides useful insights into students' perceptions of mental health support, several limitations and potential biases should be considered.

One possible limitation is **selection bias**, as students who chose to respond to the survey may already have a greater interest in mental health topics compared to the general student population.

Another potential issue is **response bias**, since participants might provide socially acceptable answers rather than completely honest responses due to the sensitive nature of mental health discussions.

Additionally, the survey relies on **self-reported data**, which may not always accurately represent actual behaviors or experiences. Participants' perceptions of stress or mental health access may vary depending on personal interpretation.

Another limitation relates to the **sample size and diversity of respondents**. If most responses come from students within the same academic program or institution, the results may not fully represent the broader student population.

Future studies could improve data reliability by increasing the number of participants, collecting responses from multiple universities, and combining surveys with other data collection methods such as interviews or system usage monitoring.

3. Systems Analysis Report

3.1 Element and Relationship Analysis

Elements:

1. Actors:
 - Students
 - Peer counselors
 - Platform admin
 - Professional counselor
 - University admin
2. Functional modules:
 - User Registration
 - Matching Assessment
 - Peer Assignment
 - Anonymous Communication
 - Appointment Scheduling
 - Resource Library

Data:

1. Inputs:
 - Registration data
 - Resource content
 - Appointment requests
 - Assessment responses (feedback)
2. Outputs
 - Assessment responses (feedback)
 - Peer matches
 - Appointment confirmations
 - Chat messages

Relations

Element 1	Element 2	Description
Students	User registration	The students create their profiles using the user registration module
Students	Matching assessment	The students answer the assessment that evaluate their profiles and needs for having a better experience
Students	Anonymous Communication	The students interact with the peer counselor or the professional counselor by the anonymous way of communication
Students	Appointment Scheduling	The students schedule an appointment with a professional counselor
Students	Resource library	The students can check the diverse material about mental health
Peer counselors	User registration	The peer counselor create their profiles using the user registration

		module
Peer counselors	Matching assessment	The peer counselors answer the assessment for identifying their abilities and capacities to help people, according to specific necessities.
Peer counselor	Resource library	The peer counselors can check the diverse material about mental health
Platform admin	User Registration	The platform admin maintain and solve possible errors in the user registration module
Platform admin	Matching Assessment	The platform admin maintain and solve possible errors in the matching Assessment module
Platform admin	Peer Assignment	The platform admin maintain and solve possible errors in the peer Assignment module
Platform admin	Anonymous Communication	The platform admin maintain and solve possible errors in the anonymous Communication module
Platform admin	Appointment Scheduling	The platform admin maintain and solve possible errors in the appointment Scheduling module
Platform admin	Resource library	The platform admin maintain and solve possible errors in the resource library module
Professional counselor	Anonymous Communication	The professional counselor interact with the students using the anonymous way of communication
Professional counselor	Appointment Scheduling	The professional counselor receive the scheduled appointments with him/her
University admin	Matching Assessment	The university admin check if the questions are collecting enough information to do the matching process
University admin	Peer Assignment	The university admin check if the matches have been according to the students necessities
University admin	Appointment Scheduling	The university admin define schedules and the professional counselor allowed to be in the app

University admin	Resource library	The university admin upload the content to the resource library
Peer Assignment	Matching assessment	The peer assignment module use the information acquired from the matching assessment module
Anonymous Communication	Peer Assignment	The anonymous communication module use the peer assignment module for doing the connections between the students and the peer counselors

3.2 Sensitivity and Variability Study

The Student Mental Health Support Platform is sensitive to several key variables that can significantly affect its performance and outcomes.

One of the most critical sensitive variables is the interaction between students and peer counselors. Unlike automated systems, peer counselors are human volunteers whose advice cannot be fully standardized or controlled. A single inappropriate or harmful comment could have a severe negative impact on a student's well-being. Since there is no way to guarantee that every peer will always give good advice, the platform is inherently sensitive to the quality of these human interactions. This makes training, supervision, and feedback mechanisms essential for risk mitigation.

Another sensitive variable is the balance between demand and supply of peer counselors. If too many students seek help at the same time and there are not enough peers available, many students may be left waiting or receive no support at all. This could discourage students from using the platform and leave their mental health needs unattended. The system is therefore highly sensitive to fluctuations in demand, especially during peak periods like midterms and finals.

Privacy and anonymity also represent a sensitive variable. According to survey data, most students consider anonymity essential for trusting the platform. If students perceive that their data is not secure—whether due to an actual breach or even rumors—they may stop using the system entirely. This makes privacy a foundational concern, as small changes in user trust can lead to large drops in adoption.

The academic calendar introduces predictable variability. During midterm and final exam periods, stress levels rise and demand for mental health support increases significantly. While this is not a small change but rather a large and expected one, the system must still be prepared to handle these surges to avoid being overwhelmed.

3.3 Complexity and Emergent Behavior

1. Nonlinear Behaviors

In linear systems, outputs are proportional to inputs. However, the platform exhibits nonlinear behaviors where small causes can produce disproportionately large effects.

- The domino effect of a single interaction: A single inappropriate comment from a peer counselor can have catastrophic effects. The affected student may not only stop using the platform but also share their negative experience with friends, discouraging dozens of potential users. This word-of-mouth effect means that one bad interaction can undo months of positive reputation building.
- The amplifier effect of positive Experiences: One exceptionally helpful interaction can trigger a virtuous chain reaction. A student who receives meaningful support may become an advocate for mental health,

encouraging friends to join the platform and even volunteering as a peer counselor themselves. This single positive interaction can ultimately bring dozens of new users and peers into the system.

- Trust Fragility: If a rumor about privacy vulnerabilities spreads—even if completely false—the resulting loss of trust can cause mass abandonment. A small piece of misinformation can generate a disproportionately large system failure.

2. Feedback Loops

The platform contains multiple feedback loops where outputs circle back to become inputs, either reinforcing growth or maintaining balance.

The first reinforcing loop involves data quality. As more students use the platform, more assessment data is collected. This data allows the matching algorithm to be trained more effectively, resulting in more accurate peer assignments. Higher accuracy leads to greater student satisfaction, which in turn encourages more students to join through recommendations. This cycle continuously improves the platform as its user base grows.

A second reinforcing loop operates through peer recruitment. When students have positive experiences, some feel grateful and motivated to give back by becoming peer counselors themselves. This increases the pool of available peers, which improves response times and support availability. Better availability leads to more positive experiences, which continues the cycle of recruitment and community growth.

3. Chaos Theory Applications

While the platform operates according to designed rules, certain aspects exhibit chaotic behavior as described in chaos theory: the system is deterministic in that it follows rules, yet practically unpredictable due to extreme sensitivity to initial conditions.

Sensitivity to Contextual Factors

Demand for mental health support cannot be predicted with precision because it depends on countless variables. For example, if an exam turns out to be harder than expected, this can trigger an unexpected demand spike that overwhelms available peers. Even something as seemingly unrelated as rainy weather can affect mood and slightly increase the number of students seeking support, though the exact magnitude remains unpredictable.

The "Butterfly Effect"

In chaos theory, the "butterfly effect" suggests that a butterfly flapping its wings could cause a tornado elsewhere. The platform exhibits similar sensitivity. A single student's positive post on social media could trigger a wave of new users. One professor giving an unexpectedly difficult exam could create a demand spike that saturates the system. A peer counselor's personal problems could affect their judgment during a critical conversation. These variations make perfect prediction impossible.

4. Unexpected Patterns

Through user interactions, several patterns have emerged that were never explicitly designed into the system. These emergent behaviors reveal both risks and opportunities that were not anticipated during the initial design phase.

Pattern 1: Risk of Exploitation

A concerning emergent pattern is the possibility of users attempting to exploit the system for inappropriate purposes. Since the platform connects students with peers anonymously, there is a risk that some individuals might join with malicious intent, seeking to take advantage of vulnerable users who may be more susceptible to manipulation. The anonymity that protects legitimate users can also be misused by those seeking to establish inappropriate relationships or exert influence over emotionally vulnerable individuals. This pattern was not considered in the original design but represents a real risk that must be addressed through moderation, monitoring, and safety protocols.

Pattern 2: Transition to Real-World Relationships

In some cases, conversations that begin as anonymous peer support may naturally evolve into genuine friendships or even romantic relationships. As trust develops through repeated interactions, students may choose to share personal contact information and continue their connection outside the platform. While this can lead to meaningful real-world friendships and expanded social support networks, it also introduces risks related to privacy and safety that were not part of the original design. The platform must balance facilitating authentic human connection with protecting user anonymity and security.

Pattern 3: Resource Spillover with External Impact

Students frequently share resources from the platform's library with friends who are not registered users. When users find articles, videos, or exercises particularly helpful, they often share them with friends or classmates outside the platform. This spillover effect extends mental health support beyond the platform's boundaries, helping students who might not have sought help directly. In some cases, these shared resources have contributed to improving the mental health of external students, creating a positive but unplanned impact on the wider student community.

3.4 Findings and Pattern Recognition

Based on the analysis of survey data collected from 25 university students, several key findings, trends, anomalies, and behavioral patterns emerged. These insights provide a data-driven foundation for understanding student mental health needs and informing the design of the proposed support platform.

Data-Driven Insights

The survey data reveals several critical insights about the current state of student mental health and attitudes toward support services:

Insight 1: High Prevalence of Mental Health Challenges

A significant majority of students (88%) reported experiencing anxiety or emotional problems during their university life. Furthermore, 72% of respondents stated that they experience academic stress "frequently" or "always." These figures confirm that mental health challenges are widespread among the student population and that there is a genuine need for accessible support services.

Insight 2: Low Utilization of Existing Services

Despite the high prevalence of mental health challenges, 68% of students have never sought help for their concerns. This disconnect between need and action suggests significant barriers to accessing care. When asked about the ease of accessing mental health services at the university, no students found it "easy" or "very easy." Instead, 44% found it "difficult" or "very difficult," and the remaining 56% responded "I don't know," indicating a profound lack of awareness about available resources.

Insight 3: Strong Interest in a Digital Platform

When asked if they would use a mental health support platform, 40% said "yes" and an additional 36% said "maybe," totaling 76% of students open to the idea. This indicates strong potential adoption for the proposed platform. Additionally, 92% of students believed that a peer-support platform could improve student mental health (60% "yes" and 32% "maybe"), validating the core concept.

Insight 4: Anonymity as a Critical Factor

A majority of students (64%) stated they would prefer anonymous support when seeking help. This finding is consistent with the low utilization of existing services, which typically require students to identify themselves. Anonymity appears to be a key factor in making students feel comfortable enough to seek support.

Insight 5: Demand for Comprehensive Support

When asked what type of support they would like to find on a mental health platform, 68% of respondents selected "all of the above" (professional counseling, peer support, and educational resources). This suggests that students want a comprehensive solution rather than a single type of support, and that the platform should offer multiple ways to meet diverse needs.

Insight 6: Openness to Peer Interaction

Regarding comfort with discussing mental health with other students, 52% responded "maybe" and 16% responded "yes," indicating that 68% are at least open to the idea of peer support. While not a strong endorsement, this suggests that peer-based approaches are not rejected and could be adopted with appropriate design and incentives.

Trend Analysis

Examination of the data across different response patterns reveals several notable trends:

Trend 1: The Action Gap

Similarly, there is a notable gap between experiencing problems (88%) and seeking help (32% have sought help). This 56-point gap suggests that even when students are aware of services, significant barriers prevent them from taking action. These barriers likely include stigma, privacy concerns, and complicated access procedures, all of which an anonymous digital platform could help address.

Trend 2: Consistency Across Support Preferences

The preference for comprehensive support (68% selected "all of the above") was consistent across respondents, regardless of their stress levels or prior help-seeking behavior. This trend indicates that students view mental health support holistically and expect a platform to offer multiple integrated options rather than isolated services.

Trend 3: Moderate but Present Peer Support Potential

While only 16% of students expressed definite comfort with peer discussions, the 52% who responded "maybe" represent a significant pool of potential users who could be converted with the right approach. This trend suggests that peer support should be positioned as one option among many, not as the only option, to avoid alienating hesitant users.

Anomaly Identification

Several responses deviated from the general patterns and warrant special attention:

Anomaly 1: The 12% Who Would Use the Platform Frequently

While most students (52%) said they would use the platform "occasionally," 12% indicated they would use it "frequently" and another 20% said "a few times per month." This smaller group of highly engaged potential users may represent students with chronic or severe needs who require more intensive support. The platform must be designed to accommodate both occasional and frequent users without overwhelming resources.

Anomaly 2: Complete Absence of "Easy" or "Very Easy" Access Responses

The fact that zero students found accessing mental health services "easy" or "very easy" is striking. Even more notable is that 56% responded "I don't know," suggesting that for a majority of students, existing services are so invisible that they cannot even rate their accessibility. This anomaly underscores the complete disconnect between institutional offerings and student awareness.

Anomaly 3: The 8% Who Do Not Believe a Platform Would Help

While 92% believed a platform could improve mental health (combining "yes" and "maybe"), 8% responded "no." Understanding why this minority is skeptical could provide valuable insights. Possible reasons include concerns about effectiveness, privacy, or a preference for face-to-face interaction.

Behavioral Patterns

Analysis of the data reveals consistent patterns in how students currently behave and how they would likely interact with a mental health platform:

Pattern 1: Silent Suffering

The data shows a clear pattern of silent suffering: high rates of mental health challenges (88%) combined with low rates of help-seeking (32%). Students are experiencing stress, anxiety, and emotional problems but are not accessing available resources. This pattern suggests that students cope alone or through informal networks rather than institutional channels.

Pattern 2: Preference for variety

The strong preference for "all of the above" (68%) indicates that students do not want to choose between different types of support. Instead, they want a platform where they can access professional counseling, peer support, and educational resources depending on their needs at any given time. This pattern suggests that the platform should integrate these options seamlessly rather than treating them as separate services.

Pattern 3: Awareness as the Primary Barrier

Across all data points, the most consistent pattern is that lack of awareness of existing services, of how to access help, of what support is available, is the primary barrier to student mental health care. The 56% who responded "I don't know" when asked about access, combined with the 68% who never sought help, paint a clear picture: students are not rejecting help; they simply do not know it exists or how to find it.

3.5 Critical Assessment

- System Performance Evaluation

The survey data reveals a clear disconnect between student needs and existing mental health services. Despite high levels of stress and emotional challenges among students, current support systems remain largely invisible and inaccessible, as evidenced by low awareness and low help-seeking rates. In contrast, the proposed digital platform aligns strongly with student preferences for anonymity, comprehensive support, and accessible design. The high level of interest expressed by students, along with their belief that such a platform could improve well-being, indicates that the proposed system would be well-received and would effectively address the gaps left by current services. These findings support the viability and potential impact of the platform as a solution to the identified problems.

- Bottleneck Identification

Once the platform is operational, several bottlenecks could emerge that would affect its ability to deliver timely and effective support to students.

The most significant bottleneck is peer counselor availability. Since peer counselors are volunteers, there is no guarantee that enough peers will be available when demand spikes, particularly during midterms and final exams. This could result in students waiting long periods for support or receiving no help at all during the moments they need it most.

Professional counselor capacity also represents a critical bottleneck. If the platform successfully identifies students requiring clinical attention, the existing university counseling services may be overwhelmed, leading to long wait times for appointments. Students with severe needs could be left without timely professional care.

Peer burnout is another potential bottleneck. If peers take on too many sessions without adequate rest or support, their mental health could suffer and the quality of their interactions could decline, creating a cycle of reduced effectiveness and higher turnover.

- Optimization Opportunities

Based on the bottlenecks identified, several opportunities exist to optimize the platform and enhance its performance once operational.

Predictive Peer Recruitment

By using the academic calendar to anticipate demand peaks, the platform could launch recruitment campaigns before exams begin, ensuring that peers are available when needed most. This would directly address the peer availability bottleneck.

Tiered Support Model

Implementing a clear escalation path self-help resources for mild stress, peer chat for moderate anxiety, and professional counseling for severe cases would ensure appropriate resource allocation and preserve professional capacity for those who need it most.

Gamified Peer Incentives

A recognition system with badges, certificates, and community acknowledgment would motivate peers to remain active and engaged, reducing turnover and mitigating peer burnout.

maintaining service availability during peak periods.

- Recommendations

Based on this critical assessment, the following recommendations are proposed for the design and development of the Student Mental Health Support Platform:

Make Visibility a Priority

The platform must be aggressively promoted through university channels, social media, and student organizations. A well-designed platform that no one knows about will fail just like the current invisible system.

Design for Anonymity First

Privacy should be the foundational design principle. All features should assume users want to remain anonymous, including anonymous registration, encrypted communications, and clear data usage policies.

Offer Multiple Support Pathways

The platform should integrate professional counseling appointments, peer chat, and self-help resources seamlessly, allowing students to move between options based on their needs.

Build a Peer Program from the Start

A structured peer counselor program should be developed alongside the platform, including recruitment, training, supervision, and retention strategies to ensure a stable and capable peer workforce.

Implement Predictive Capacity Planning

Use academic calendar data to anticipate demand peaks during exams and midterms, recruiting and scheduling additional peers during these periods.

Design for Scalability

Implement auto-scaling infrastructure from the beginning to ensure the platform can handle demand spikes without performance degradation.

4. Visual System Representation

4.1 System Architecture Diagram:

The system architecture is organized into three main layers: user, application, and infrastructure, following a modular, scalable, and privacy-focused approach. This structure allows for a clear visualization of how users interact with the functional modules and how the system's operation is technically supported.

1. User Layer: This layer represents the users who interact directly with the platform:

- Students: access via web or mobile, create anonymous profiles, take assessments, use anonymous chat, schedule appointments, and consult educational resources.
- Peer Counselors: trained volunteers who offer emotional support through anonymous communication, take skills assessments, and access the resource library.
- Professional Counselors: receive appointment requests, interact via anonymous chat, and provide specialized guidance.
- University Administration: oversees policy compliance, defines service hours, and uploads content to the library.
- Platform Administrators: operate and maintain the functional modules, ensuring system availability and security.

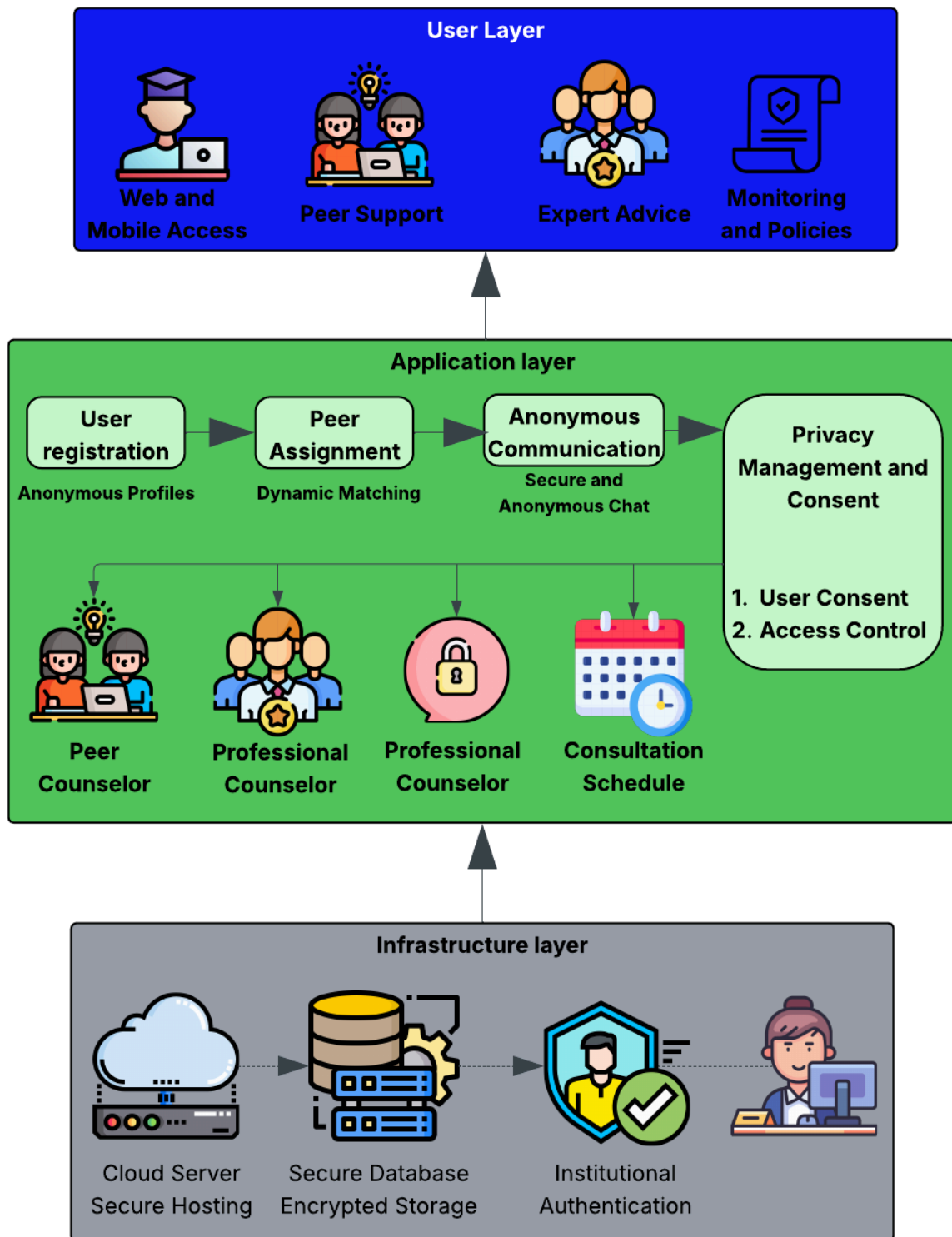
2. Application Layer: This layer contains the functional modules that enable the system's operation:

- User Registration: Allows the creation of anonymous, institutionally authenticated profiles.
- Peer Matching: Performs dynamic matching between students and peer counselors.
- Anonymous Communication: Enables secure chat between users, preserving anonymity.
- Appointment Scheduling: Allows scheduling sessions with professional counselors.
- Privacy and Consent Management: A cross-cutting module that guarantees informed consent and access control.

3. Infrastructure Layer: This technological layer supports the secure and continuous operation of the system:

- Cloud Server: Hosts the platform, ensuring availability and scalability.
- Secure Database: Stores anonymous profiles, appointments, and assessments, excluding medical records.
- Institutional Authentication: Validates access using university credentials.

Layered architecture diagram:



4.2 Process Flow / Relationship Map

The diagram represents the journey a student takes within the platform, from initial registration to the feedback that drives continuous system improvement.

1. Process Start: The student accesses the platform and completes the registration and initial questionnaire, which gathers their needs and preferences.

2. Decision on Anonymity: The system asks the question: “Do you prefer anonymous support?”

- If the answer is Yes → the student proceeds to the anonymous matching module, which connects them with a peer counselor or a professional counselor.
- If the answer is No → the student is referred directly to a professional counselor.

3. Support Session: Once assigned, the student participates in a support or counseling session, which may include:

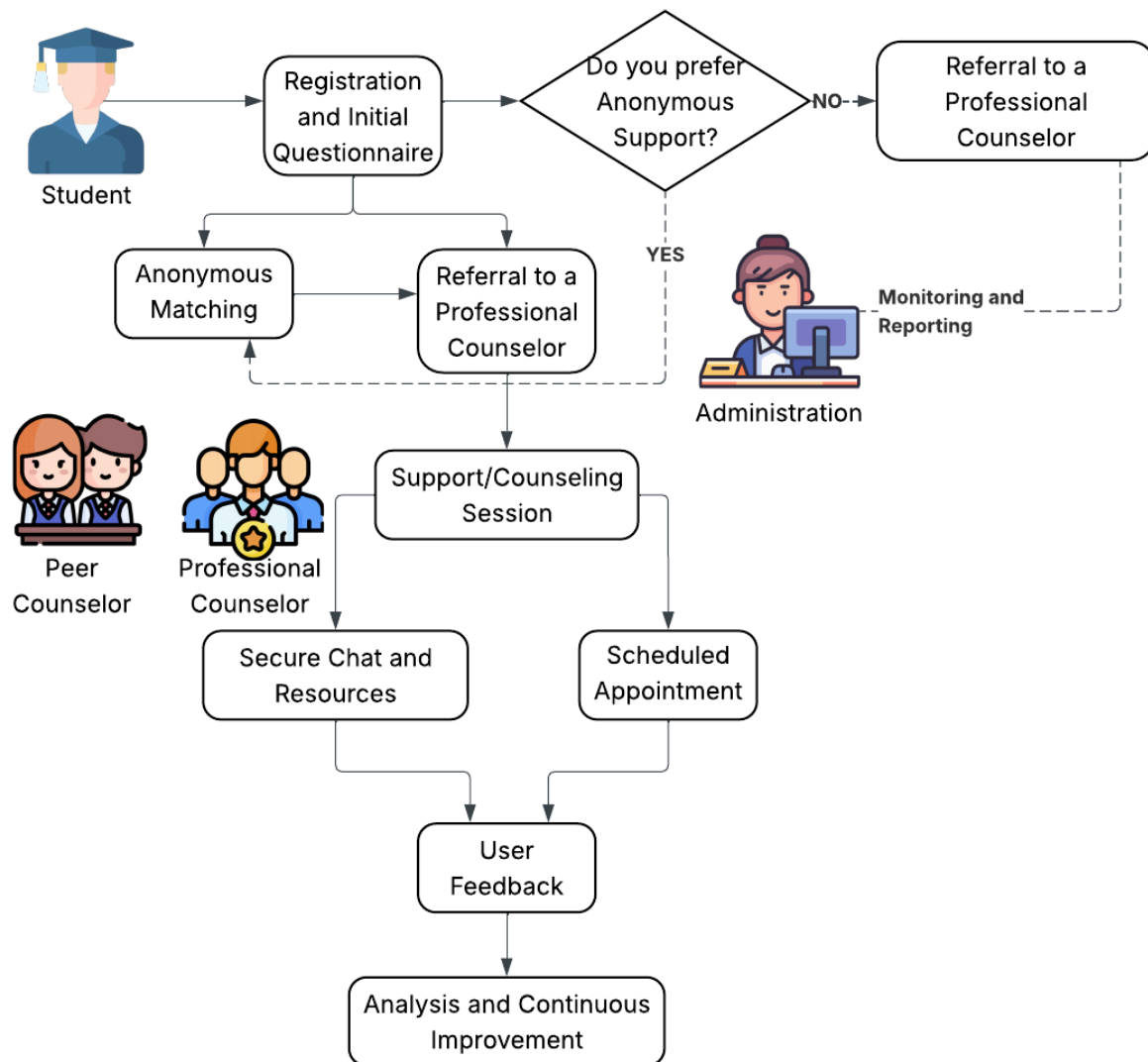
- Secure chat and educational resources (when immediate support is needed).
- A scheduled appointment (when more structured attention with a professional is required).

4. Feedback: At the end of the session, the student provides user feedback, which is recorded in the system.

5. Continuous Improvement: Feedback feeds into the analysis and continuous improvement module, which adjusts processes, optimizes matching, and strengthens service quality.

6. Administrative Oversight: University administration participates through monitoring and reporting, ensuring policy compliance, data security, and service quality.

Process flowchart and relationships:



5. Documentation and Repository Management

5.1 Repository Structure

A dedicated GitHub repository has been created for this course. The repository follows the folder structure required for all workshop deliverables across the semester:

Folder	Contents
/data	Raw data files, survey responses, and collected measurements
/diagrams	System architecture diagrams and process flow maps
/docs	Final PDF report and supporting academic documents
/analysis	Analysis scripts, tools, and methodology documentation
/README.md	Project overview, team members, data collection process, key findings

5.2 Repository Link

The complete project repository, including all raw data, diagrams, scripts, and documentation, is available at:

https://github.com/Alan05ss/Mental_Health_Project.git

5.3 README Overview

The repository README.md file documents the following:

- Project title and team member names with assigned roles
- System description and scope summary
- Data collection methodology and process
- Key findings from the systems analysis

References

Sierra, C. A. (2026). Workshop No. 1 — Systems Analysis [Course assignment]. Universidad Distrital Francisco José de Caldas.